Problem 11 | Problems T5_MechanicalProperties.pdf #11 | Hooke's law · Poisson's ratio

Elastic Deformation of a 15 mm Steel Cylinder under 50 kN

Original statement · Problems T5_MechanicalProperties.pdf, p. 4

1. *A cylindrical steel sample with a diameter of 15 mm and a length of 250 mm experiences elastic deformation when a load of 50 kN is applied. Considering that for this steel E = 207 GPa and ν = 0.27, determine the elongation in the direction of the force applied and the change in diameter of the sample.*

Official Solution: $\Delta l = 0.342$ mm, $\Delta\phi = -0.0055$ mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Elastic deformation.** The statement says that the load produces only elastic deformation, so Hooke's law holds, $\sigma = E\varepsilon$ [Session 11 Slide 13].
2. **Uniaxial, uniform stress** along the axis of the cylinder: $\sigma = F/A_0$ and $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
3. **Isotropic material.** With a uniaxial stress, the transverse strain is $\varepsilon_{lat} = -\nu\,\varepsilon_{ax}$, equal in the two transverse directions [Session 11 Slide 19].
4. The cross-section is circular and uniform, so that the diameter changes in the same proportion in every direction of the section.

Data

Quantity	Symbol	Value	In SI
Load	F	50 000 N	50 000 N
Diameter	d_0	15.0 mm	15.0×10^{-3} m
Length	l_0	250 mm	250×10^{-3} m
Young's modulus	E	207 GPa	207 000 N/mm²
Poisson's ratio	ν	0.27	

Unknowns: the elongation Δl in the direction of the force and the change of the diameter Δd.

PHASE 2 · FRAMES & STRATEGY

One direction for the load (the axis) and the two equal transverse directions of the cylinder; no change of basis.

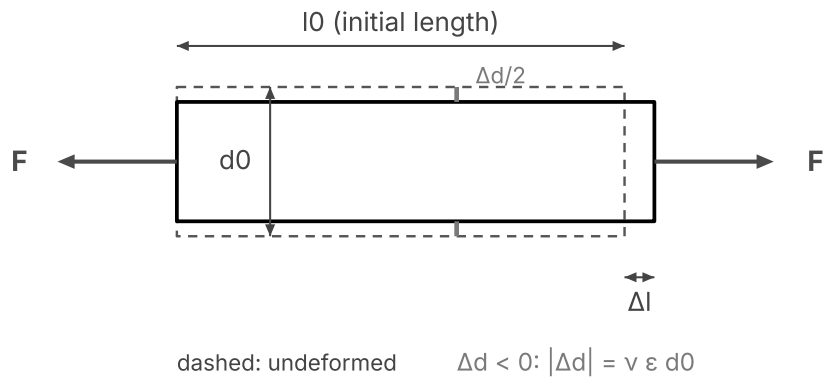


Figure 10. Cylindrical bar under an axial load F: it elongates by Δl and its diameter shrinks by Δd (schematic, exaggerated).

Strategy. The elongation follows from the stress and the modulus, and the contraction of the diameter follows from the same strain through Poisson's ratio [Session 11 Slide 19]:

$$\sigma = \frac{F}{A_0} \qquad \varepsilon_{ax} = \frac{\sigma}{E} \qquad \Delta l = \varepsilon_{ax}\, l_0 \qquad \varepsilon_{lat} = -\nu\, \varepsilon_{ax} \qquad \Delta d = \varepsilon_{lat}\, d_0$$

The minus sign says that the diameter decreases when the sample is stretched.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15.0\ \text{mm})^2}{4}$$

The square is $15.0^2 = 225.00$, then $\pi \times 225.00 = 706.86$ and dividing by 4:

$$A_0 = 176.71\ \text{mm}^2$$

2. Stress and axial strain

$$\sigma = \frac{F}{A_0} = \frac{50\,000\ \text{N}}{176.71\ \text{mm}^2} = 282.94\ \text{MPa}$$

$$\varepsilon_{ax} = \frac{\sigma}{E} = \frac{282.94\ \text{N/mm}^2}{207000\ \text{N/mm}^2} = 1.3669 \times 10^{-3}$$

3. Elongation in the direction of the force

$$\Delta l = \varepsilon_{ax}\, l_0 = 1.3669 \times 10^{-3} \times 250\ \text{mm} = \mathbf{0.342\ mm}$$

4. Change of the diameter

The transverse strain is

$$\varepsilon_{lat} = -\nu\, \varepsilon_{ax} = -0.27 \times 1.3669 \times 10^{-3} = -3.6905 \times 10^{-4}$$

and the change of the diameter is

$$\Delta d = \varepsilon_{lat}\, d_0 = -3.6905 \times 10^{-4} \times 15.0\ \text{mm} = \mathbf{-0.00554\ mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** (N/mm²)/(N/mm²) has no units for the strains, and strain × mm gives mm for Δl and Δd.
- **Elastic range.** $\varepsilon_{ax} = 0.1367$ %, a small strain of the order of those of the elastic deformation of a metal, consistent with the statement.
- **Volume.** The relative change of volume is $\Delta V/V \approx \varepsilon_{ax} + 2\varepsilon_{lat} = \varepsilon_{ax}(1 - 2\nu) = 6.288 \times 10^{-4}$, positive: a stretched elastic body increases its volume since $\nu < 0.5$ (a perfectly incompressible material, $\nu = 0.5$, would give zero) [Session 11 Slide 19].
- **Sign.** $\Delta l > 0$ and $\Delta d < 0$: the sample gets longer and thinner.

Interpretation

The sample elongates by 0.342 mm, about a third of a millimetre in 250 mm, and the diameter shrinks by only 5.54 micrometres. The contraction is a fraction $\nu = 0.27$ of the relative elongation, which is why it is so small and why it is difficult to measure directly.

Check against the official solution

Item	Official	Computed	Comment
Δl	0.342 mm	0.342 mm	agrees
Δd	−0.0055 mm	-0.0055 mm	agrees